厦门兰阶教育 2022 年学科奥赛•物理珍题集训
模拟试卷三（卷Ⅰ）　命题者：苏国珍教授

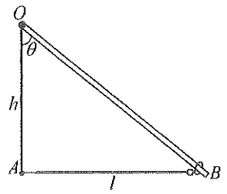
学　校_____ 姓　名_____	条码粘贴处

请在各题目的答题区域内作答，超出黑色矩形边框限定区域的答案无效

一.（36 分）如图所示，直杆 OA 与 OB 固定于竖直面内，两者成夹角 $\theta\ (0<\theta<\pi/2)$。一细绳子一端固定于 A 点，另一端与套在杆 OB 上的小环相连，一质量为 m 的小珠子穿在绳子上。已知 $\overline{OA}=h$，绳子的长度 $l=h\tan\theta$。初始绳子刚好拉直，小珠子位于紧靠小环的位置。系统从此位置静止释放。设细绳和小环的质量均可忽略，小环与杆之间以及小珠子与绳子之间的摩擦力均可忽略。试求：

（1）小珠子到达最低点时的速度大小；[12 分]

（2）当小珠子位于最低点时，绳子的张力。[24 分]



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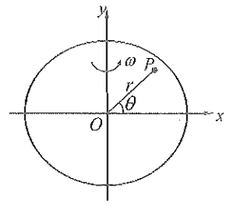
二.（40 分）一颗质量为 M 的星球，由不可压缩的流体组成，该星球以恒定的角速度 ω 缓慢自转。由于转动，其赤道处比两极处略有凸起。已知 $\omega=0$ 时，星球的半径为 R，忽略其它星球对该星球的作用。作为一级近似，以下问题的结果均保留到 ω^2 项，且在计算万有引力时，可将星球视为质量均匀分布的球体。

（1）设赤道处和两极处的半径（表面到星球中心的距离）分别记为 R_e 和 R_p，试求 R_e-R_p 的表达式（结果用 M、ω、R 和物理常量表示）；[12 分]

（2）若地球可视为由不可压缩的流体组成，半径 $R=6.37\times10^3\ \mathrm{km}$，质量 $M=5.96\times10^{24}\ \mathrm{kg}$，万有引力常量取 $G=6.67\times10^{-11}\ \mathrm{N\cdot m^2/kg^2}$，试估算地球的 R_e-R_p 的值；[6 分]

（3）若星球内某一点 P 的位置用极坐标 (r,θ) 表示，试求该点压强 $p(r,\theta)$ 的函数表达式。（结果用 M、ω、R 和物理常量表示）[22 分]



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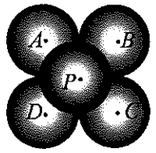
三.（44 分）4 个相同的球壳 A、B、C、D 靠在一起放在水平面上，球壳 P 放在 4 个球壳的上方。已知 5 个球壳的半径均为 R，球壳 P 的质量为 $4m$，其余 4 个球壳的质量均为 m，所有球壳质量均匀分布。先扶住 5 个球壳，使系统保持静止，然后轻轻放开。设重力加速度为 g。

（1）假设放开后系统仍能保持平衡状态，则 P 与下面 4 个球壳之间以及下面 4 个球壳与水平面之间的最小摩擦系数分别是多少？（假设球壳 P 与其它各球壳之间的摩擦系数相同，各球壳与水平面之间的摩擦系数也相同）[10 分]

（2）假设各球壳之间以及各球壳与水平面之间均无摩擦。放开后系统在重力作用下从静止开始运动。求球壳 P 碰到水平面时的速度。 [14 分]

（3）假设球壳与水平面之间无摩擦，各球壳之间的摩擦系数足够大，则球壳 P 碰到水平面时的速度是多少？[20 分]

已知质量为 m，半径为 R 的匀质球壳绕其直径的转动惯量为：$J = 2mR^2/3$ 。



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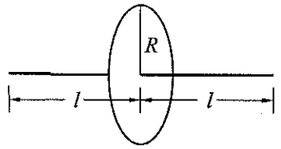
四.（40 分）如图，有一均匀带电直线，长度为 $2l$，所带总电量为 q，在空间位置保持固定。另一均匀带电圆环，半径为 R，质量为 m，所带电量为 Q，可在空间运动。系统的初态为：圆环的轴线与带电直线重合，中心位于直线的中点，系统处于静态平衡状态。

（1）设圆环只能沿轴线作平动。试讨论圆环的平衡性质；若为稳定平衡，求圆环在平衡位置附近做微小振动的周期；[10 分]

（2）设圆环只能沿垂直于轴线的某一方向作平动。试讨论圆环的平衡性质；若为稳定平衡，求圆环在平衡位置附近做微小振动的周期；[15 分]

（3）设圆环只能绕与它的直径重合的某一固定轴作定轴转动，且带电直线的长度 $2l >> R$ 。试讨论圆环的平衡性质；若为稳定平衡，求圆环在平衡位置附近做微小振动的周期。[15 分]



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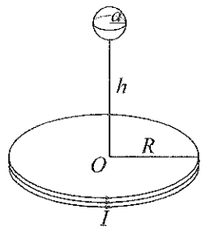
五.（40 分）利用磁力的作用可以产生磁悬浮的效果。如图，在水平面上固定一个由 N 匝绝缘导线重叠绕制成的线圈，其半径为 R，每匝通以逆时针方向（从上方看）的恒定电流 I。在线圈中心上方 h 处有一个半径 $a << R$ 的薄球壳，球壳所带电量为 Q，质量为 m，且电量和质量都均匀分布。已知当球壳在该处以角速度 ω_0 绕过球心的竖直轴转动时，球壳刚好能处于悬浮状态。设重力加速度为 g，球壳上的电荷产生的磁场可忽略。

（1）求 ω_0 的大小，并确定其方向；[20 分]

（2）当球壳处于悬浮状态时，突然切断线圈中的电流。求切断电流后瞬间，球壳的角速度。设切断后，线圈中的电流在极短的时间内减到零。已知球壳绕其直径的转动惯量为 $J = 2mR^2/3$。[20 分]



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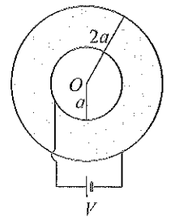
六.（40 分）如图，两同轴导体圆柱壳，半径分别为 a 和 $2a$，长均为 l $(l >> a)$，在两圆柱壳之间充满双原子分子理想气体，总分子数为 N。内外圆柱壳分别与电压为 V 的恒压源的正负极相连。由于电场的作用，气体分子被极化，达到极化平衡时，分子的电偶极矩 $\vec{p}_e = \gamma\vec{E}$，其中 $\vec{E}$ 为分子所在处的外电场强度，γ 为与分子结构有关的常量。忽略分子电偶极矩产生的电场，忽略重力的作用。

（1）设某一气体分子到圆柱轴线的距离为 r，求该分子受到的电场力；[12 分]

（2）设系统处于温度为 T 的平衡态，求两圆柱壳之间气体分子数密度的分布；[14 分]

（3）求该系统的定容热容与温度 T 的关系。[14 分]



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七.（36 分）军用无线电技术利用干涉来保证通信的安全性。某指挥部的无线电发射基地设在 A 处，B、C 为两不同作战单位，可接收无线电信号。为防止泄密，指挥部希望与其中一个作战单位通信时，另一个作战单位不能同时收到信号。为此指挥部在发射基地架设了两根发射天线 S_1 和 S_2，它们可以同时朝各个方向发射固定波长 λ 的无线电信号，两天线所发出信号的相位差可在每次发射前调整，如图 1 所示。设 S_1、S_2 之间的距离为 d，AB 与 AC 的夹角为 $\varphi\ (0 \le \varphi \le \pi)$，$A$ 到 B、C 的距离远大于 d。S_1、S_2 可绕 A 旋转，若用 α 表示 S_1S_2 转到 AB 转过的角度（转动方向为逆时针时，$\alpha > 0$，顺时针时，$\alpha < 0$），则 α 可在 $-\pi < \alpha < \pi$ 范围内变化。

（1）为了使一个作战单位收到最强信号时，另一个作战单位收不到信号，d 的最小值 $d_{\min} = ?$ 此时对应的 $\alpha = ?$ [14 分]

（2）若 $d = d_{\min}$，则为使 B 收到最强信号，而 C 收不到信号，S_1、S_2 之间信号的相位差 $\Delta\varphi_0$ 是多少？[10 分]

（3）为了加强接收信号，指挥部在发射基地架设了 N 根发射天线 S_1、S_2、…、S_N $(N > 2)$，它们可以同时朝各个方向发射波长为 λ 的信号，如图 2 所示。设 N 根天线沿直线排列，组成一维天线阵列，相邻天线之间的距离为 d，相邻天线发出的信号具有相同的位相差，该天线阵列可绕 A 旋转，即角度 α 在 $-\pi < \alpha < \pi$ 范围内可调。试求为了使一个作战单位收到最强信号时，另一个作战单位收到的信号强度为零，d 的最小值 $d_{\min} = ?$ 此时对应的 $\alpha = ?$ [12 分]

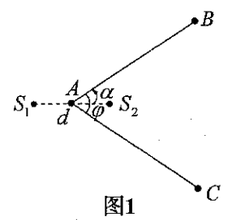

图1

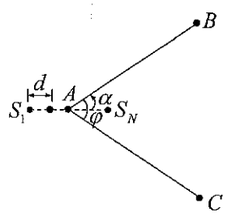

图2

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八.（44 分）某种原子 H 内部只有两个能级，低能级的称基态，高能级的称激发态，激发态与基态的能量差为 E_0，原子处于基态时，静质量为 m_0。处于激发态的原子是不稳定的，当它们跃迁到基态时，将对外发出光子。已知在与原子相对静止的参照系 S' 中测，原子 H 单位时间内从激发态跃迁的基态的概率为 λ，跃迁时沿各个方向等概率地发出光子。今有一批 H 原子，总原子数为 N_0 $(N_0 >> 1)$，当实验室参照系 S 和参照系 S' 系中的钟均指在零 $(t_0 = t_0' = 0)$ 时，N_0 个原子均处于激发态，此时 N_0 个原子相对于实验室参照系 S 具有沿 $+x$ 方向的共同速度 v。

设 $m_0 >> E_0 / c^2$，这种情况下发光过程中原子的缓冲可忽略不计，请在该条件下解答以下问题：

（1）在 S' 系中测，任意 t' 时刻，这一批原子发光的总功率；[8 分]

（2）在 S 系中测，任意 t 时刻，这一批原子发光的总功率；[8 分]

（3）在 S 系中测，这一批原子发光功率按方向的分布（求出沿与 x 轴夹角为任意 θ $(0 \le \theta \le \pi)$ 方向单位立体角的发光功率）；[12 分]

设 $m_0 >> E_0 / c^2$ 的条件不满足，这种情况下发光过程中原子的缓冲不可忽略，请在该条件下解答以下问题：

（4）在 S 系中测，经过足够长时间后（几乎所有原子都跃迁到了基态），N_0 个 H 原子发出的光子的总能量和总动量。[16 分]

本题所有问题均需考虑相对论效应。

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中学生物理奥赛培训

模拟试卷（三）解答

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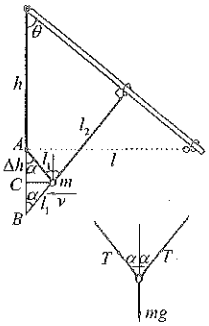
(gzsu@xmu.edu.cn)

福建 厦门 2022.6

题一.

(1) 最低点，重力势能最小；
最低点，为小球平衡位置。



$$\alpha=\frac{\pi}{2}-\theta$$

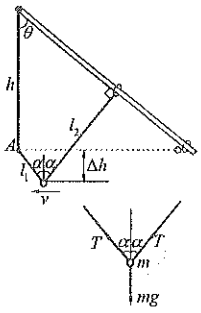
$$2\Delta h+h=\frac{l_1+l_2}{\sin\theta}=\frac{l}{\sin\theta}=\frac{h}{\cos\theta}\qquad (l=h\tan\theta)$$

$$\Delta h=\frac{1}{2}h(\frac{1}{\cos\theta}-1)$$

$$\frac{1}{2}mv^2=mg\Delta h\Rightarrow$$

$$v=\sqrt{2g\Delta h}=\sqrt{gh(\frac{1}{\cos\theta}-1)}$$

(2)



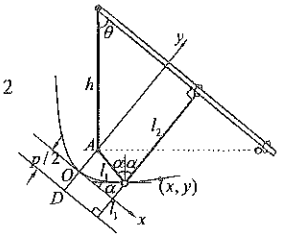
$$2T\cos\alpha-mg=\frac{mv^2}{\rho}=\frac{2mg\Delta h}{\rho}$$

$$T=mg(\frac{1}{2\cos\alpha}+\frac{\Delta h}{\rho\cos\alpha})$$

$$\rho\cos\alpha=?$$

$$v=\sqrt{2g\Delta h},\Delta h=h(\frac{1}{\cos\theta}-1)$$

小球的运动轨道为抛物线；



$$2\overline{OA}+h\sin\theta=l\Rightarrow$$

$$\overline{OA}=\overline{OD}=p/2=(l-h\sin\theta)/2$$

$$x^2=2py=2(l-h\sin\theta)y$$

$$\rho=(1+y'^2)^{3/2}/y''$$

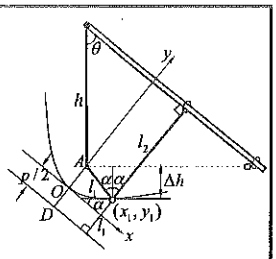
$$\tan\alpha=y'\Rightarrow\cos\alpha=1/\sqrt{1+y'^2}$$

$$\rho\cos\alpha=(1+y'^2)/y''$$

$$y=\frac{x^2}{2(l-h\sin\theta)}$$

$$y'=\frac{x}{l-h\sin\theta},\ y'^2=\frac{x^2}{(l-h\sin\theta)^2}=\frac{2y}{l-h\sin\theta},\ y''=\frac{1}{l-h\sin\theta}$$

$$\rho\cos\alpha=l-h\sin\theta+2y$$



$$y_1=l_1-\frac{p}{2}=l_1-\frac{1}{2}(l-h\sin\theta)$$

$$\rho\cos\alpha=2l_1$$

$$T=mg(\frac{1}{2\cos\alpha}+\frac{\Delta h}{2l_1})$$

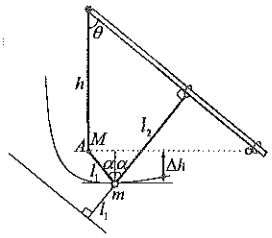
$$=\frac{1}{2}mg(\frac{1}{\cos\alpha}+\cos\alpha)$$

$$=\frac{1}{2}mg(\frac{1}{\sin\theta}+\sin\theta)$$

$$\rho\cos\alpha=l-h\sin\theta+2y,T=mg(\frac{1}{2\cos\alpha}+\frac{\Delta h}{\rho\cos\alpha}),\alpha=\frac{\pi}{2}-\theta,\frac{p}{2}=\frac{1}{2}(l-h\sin\theta)$$

另解：

小球的运动轨道为抛物线；
设想该抛物线为质量为 m 的质点在固定于 A 点的质量为 M 的质点的万有引力作用下的运动轨道。



$$\begin{cases}\dfrac{GmM}{l_1^2}\cos\alpha=\dfrac{mv^2}{\rho}\\ -\dfrac{GmM}{l_1}+\dfrac{1}{2}mv^2=0\end{cases}$$

$$\Rightarrow\rho\cos\alpha=2l_1$$

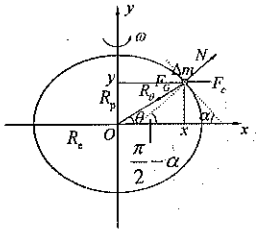
$$T=mg(\frac{1}{2\cos\alpha}+\frac{\Delta h}{2l_1})=\frac{1}{2}mg(\frac{1}{\cos\alpha}+\cos\alpha)=\frac{1}{2}mg(\frac{1}{\sin\theta}+\sin\theta)$$

$$T=mg(\frac{1}{2\cos\alpha}+\frac{\Delta h}{\rho\cos\alpha}),\alpha=\frac{\pi}{2}-\theta$$

题二

(1.1)

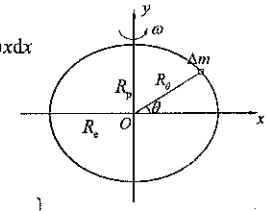
$$\begin{cases} F_t = F_G\sin(\frac{\pi}{2}-\alpha-\theta) - F_c\cos\alpha = 0 \\ F_G \approx G\frac{M\Delta m}{R_\theta^2} \\ F_c = \Delta m\omega^2 x = \Delta m\omega^2 R_\theta\cos\theta \end{cases}$$



$$\frac{\cos(\alpha+\theta)}{\cos\theta\cos\alpha} = \frac{\omega^2 R_\theta^3}{GM}$$

$$\left.\begin{aligned} &1-\tan\theta\tan\alpha = \frac{\omega^2 R_\theta^3}{GM} \\ &\tan\theta = \frac{y}{x},\ \tan\alpha = -\frac{dy}{dx} \end{aligned}\right\} \Rightarrow \frac{ydy}{xdx} = \frac{\omega^2 R_\theta^3}{GM} - 1$$

$$ydy = (\frac{\omega^2 R_\theta^3}{GM}-1)xdx \approx (\frac{\omega^2 R^3}{GM}-1)xdx$$



$$\int ydy = \int(\frac{\omega^2 R^3}{GM}-1)xdx$$

$$y^2 = -(1-\frac{\omega^2 R^3}{GM})x^2 + C$$

$$\left.\begin{aligned} &x=0, y=R_p \Rightarrow R_p^2 = C \\ &x=R_e, y=0 \Rightarrow 0 = -(1-\frac{\omega^2 R^3}{GM})R_e^2 + C \end{aligned}\right\} \Rightarrow R_p^2 = (1-\frac{\omega^2 R^3}{GM})R_e^2$$

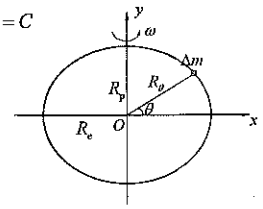
$$R_p = R_e\sqrt{1-\frac{\omega^2 R^3}{GM}} \approx R_e(1-\frac{\omega^2 R^3}{2GM})$$

$$R_e - R_p = \frac{\omega^2 R^3 R_e}{2GM} \approx \frac{\omega^2 R^4}{2GM}$$

$$\frac{ydy}{xdx} = \frac{\omega^2 r^3}{GM} - 1$$

另解：星球表面为等势能面。

$$U \approx -G\frac{M\Delta m}{R_\theta} - \frac{1}{2}\Delta m\omega^2(R_\theta\cos\theta)^2 = C$$



$$\frac{GM}{R_\theta} + \frac{1}{2}\omega^2 R_\theta^2\cos^2\theta = C'$$

$$R_\theta|_{\theta=\pi/2} = R_p \Rightarrow C' = \frac{GM}{R_p}$$

$$R_\theta \approx R_p(1+\frac{\omega^2 R_\theta^3\cos^2\theta}{2GM})$$

$$R_\theta|_{\theta=0} = R_e \Rightarrow R_e = R_p(1+\frac{\omega^2 R_e^3}{2GM})$$

$$R_e - R_p = \frac{\omega^2 R_e^3 R_p}{2GM} \approx \frac{\omega^2 R^4}{2GM}$$

(2)

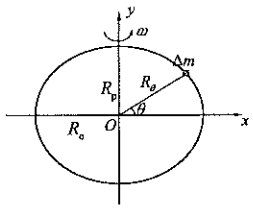
$R = 6.37\times10^3$ km

$M = 5.96\times10^{24}$ kg

$\omega = \frac{2\pi}{T} = \frac{2\pi}{24\times3600} \approx 7.27\times10^{-5}\ s^{-1}$

$G = 6.67\times10^{-11}\ N\cdot m^2/kg^2$

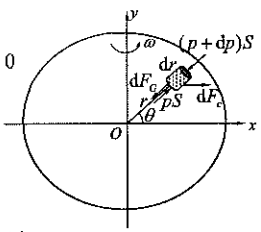
$R_e - R_p = \frac{\omega^2 R^4}{2GM}M \approx 11.0$ km



$$R_e - R_p = \frac{\omega^2 R^4}{2GM}$$

(3)

$$\begin{cases} pS + dF_c\cos\theta - (p+dp)S - dF_G = 0 \\ dF_G \approx G\frac{Mrdm}{R^3} \\ F_c = dm\omega^2 r\cos\theta \\ dm = \rho Sdr\ \ (\rho = \frac{3M}{4\pi R^3}) \end{cases}$$



$$dp = \rho(-\frac{GM}{R^3} + \omega^2\cos^2\theta)rdr\ \ (\theta\text{恒定})$$

$$\int_{p(r,\theta)}^{0} dp = \int_r^{R_\theta}\rho(-\frac{GM}{R^3} + \omega^2\cos^2\theta)rdr$$

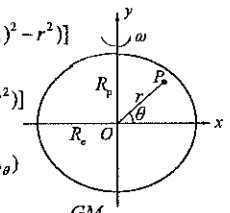
$$p(r,\theta) = \frac{1}{2}\rho(\frac{GM}{R^3} - \omega^2\cos^2\theta)(R_\theta^2 - r^2)$$

$$R_\theta = R_p(1+\Delta_\theta)\ \ (\Delta_\theta = \frac{\omega^2 R_\theta^3\cos^2\theta}{2GM} \approx \frac{\omega^2 R^3\cos^2\theta}{2GM})$$

$$p(r,\theta) = \frac{1}{2}\rho(\frac{GM}{R^3} - \omega^2\cos^2\theta)[R_p^2(1+\Delta_\theta)^2 - r^2)]$$

$$\approx \frac{1}{2}\rho(\frac{GM}{R^3} - \omega^2\cos^2\theta)[R_p^2(1+2\Delta_\theta) - r^2)]$$

$$\approx \frac{1}{2}\rho(\frac{GM}{R^3} - \omega^2\cos^2\theta)(R_p^2 - r^2 + 2R^2\Delta_\theta)$$



$$\approx \frac{1}{2}\rho[\frac{GM}{R^3}(R_p^2 - r^2) - \omega^2(R^2 - r^2)\cos^2\theta + 2\Delta_\theta\frac{GM}{R}]$$

$$= \frac{1}{2}\rho[\frac{GM}{R^3}(R_p^2 - r^2) + \omega^2 r^2\cos^2\theta]\ (\text{保留到}\omega^2\text{项})$$

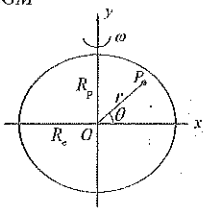
$$p(r,\theta) = \frac{1}{2}\rho(\frac{GM}{R^3} - \omega^2\cos^2\theta)(R_\theta^2 - r^2),\ R_\theta = R_p(1+\frac{\omega^2 R_\theta^3\cos^2\theta}{2GM})$$

$$\begin{cases} R_e = R_\theta\big|_{\theta=0} = R_p(1+\Delta_0)(\Delta_0 = \dfrac{\omega^2 R_e^3}{2GM} \approx \dfrac{\omega^2 R^3}{2GM}) \\ \dfrac{4\pi}{3} R_e^2 R_p = \dfrac{4\pi}{3} R^3 \end{cases}$$

$$R_p^3(1+\Delta_0)^2 = R^3$$

$$R_p^2 = R^2(1+\Delta_0)^{-4/3} \approx R^2(1-\frac{4}{3}\Delta_0)$$

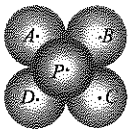
$$\approx R^2(1-\frac{2\omega^2 R^3}{3GM})$$



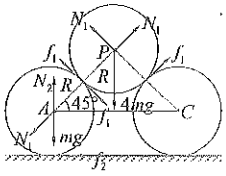
$$p(r,\theta) = \frac{1}{2}\rho[\frac{GM}{R^3}(R^2-r^2) - \omega^2(\frac{2}{3}R^2 - r^2\cos^2\theta)] \quad (\rho = \frac{3M}{4\pi R^3})$$

$$p(r,\theta) = \frac{1}{2}\rho[\frac{GM}{R^3}(R_p^2-r^2) + \omega^2 r^2\cos^2\theta]$$

题三.



(1)
$$\begin{cases} 4N_1/\sqrt{2} + 4f_1/\sqrt{2} = 4mg \\ f_1/\sqrt{2} + f_2 = N_1/\sqrt{2} \\ N_2 = N_1/\sqrt{2} + f_1/\sqrt{2} + mg \\ f_1 R = f_2 R \end{cases}$$



$f_1 = (\sqrt{2}-1)mg$

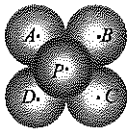
$N_1 = mg$

$f_2 = (\sqrt{2}-1)mg$

$N_2 = 2mg$

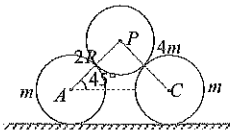
$$\mu_{1,\min} = \frac{f_1}{N_1} = \sqrt{2}-1,\ \mu_{2,\min} = \frac{f_2}{N_2} = \frac{\sqrt{2}-1}{2}$$

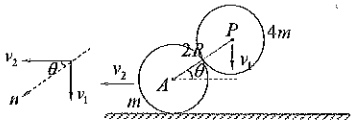
(2)



$$\begin{cases} \dfrac{1}{2}(4m)v_1^2 + 4\times\dfrac{1}{2}mv_2^2 + 4mg\cdot 2R\sin\theta \\ = 4mg\cdot 2R\sin 45^\circ \\ v_1\sin\theta = v_2\cos\theta \end{cases}$$

$$\begin{cases} v_1 = \sqrt{2gR(\sqrt{2}-2\sin\theta)}\cos\theta \\ v_2 = \sqrt{2gR(\sqrt{2}-2\sin\theta)}\sin\theta \end{cases}$$

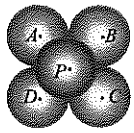




*P*脱离*A*的位置：

*P*脱离时，*P*对*A*的压力为零。

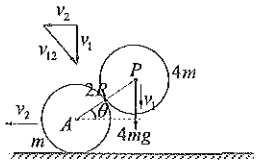
以*A*为参照系（脱离此时为惯性参照系）：



$$4mg\sin\theta = 4m\frac{v_{12}^2}{2R} = 4m\frac{v_1^2+v_2^2}{2R}$$

$$= 4mg(\sqrt{2}-2\sin\theta)$$

$$\Rightarrow \sin\theta = \frac{\sqrt{2}}{3}$$



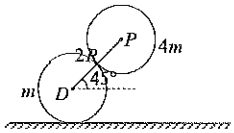
$$v_1 = \sqrt{2gR(\sqrt{2}-2\sin\theta)}\cos\theta,\ v_2 = \sqrt{2gR(\sqrt{2}-2\sin\theta)}\sin\theta$$

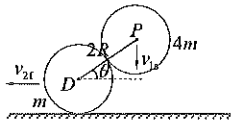
*P*碰到水平面时速度：

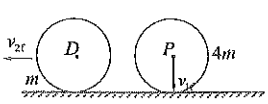
$$v_{1s} \equiv v_1\big|_{\sin\theta=\frac{\sqrt{2}}{3}} = \frac{1}{3}\sqrt{\frac{14\sqrt{2}}{3}gR}$$

$$v_{1f} = \sqrt{v_{1s}^2 + 4gR\sin\theta}$$

$$v_{1f} = \frac{5\times 2^{3/4}}{3^{3/2}}\sqrt{gR} \approx 1.62\sqrt{gR}$$



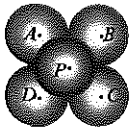




$v_1 = \sqrt{2gR(\sqrt{2}-2\sin\theta)}\cos\theta$

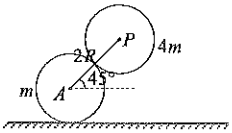
脱离位置：$\sin\theta = \sqrt{2}/3$

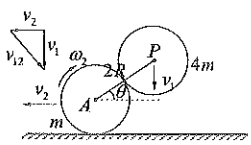
(3)



$$\begin{cases} \dfrac{1}{2}(4m)v_1^2 + 4\times\dfrac{1}{2}mv_2^2 + 4\times\dfrac{1}{2}J\omega_2^2 + 4mg\cdot 2R\sin\theta \\ = 4mg\cdot 2R\sin 45^\circ \\ v_1\sin\theta = v_2\cos\theta \\ v_{12} = \sqrt{v_1^2+v_2^2} = R\omega_2,\ J = \dfrac{2}{3}mR^2 \end{cases}$$

$$\begin{cases} v_1 = \sqrt{\dfrac{6gR(\sqrt{2}-2\sin\theta)}{5}}\cos\theta \\ v_2 = \sqrt{\dfrac{6gR(\sqrt{2}-2\sin\theta)}{5}}\sin\theta \end{cases}$$

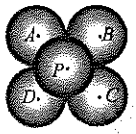




P脱离A的位置：

P脱离时，P对A的压力为零。

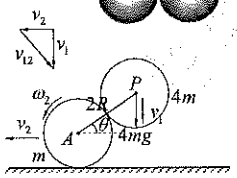
以A为参照系（脱离此时为惯性参照系）：



$$4mg\sin\theta = 4m\frac{v_1'^2}{2R} = 4m\frac{v_1^2+v_2^2}{2R}$$

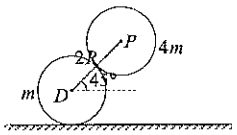
$$= 4m\frac{3g(\sqrt{2}-2\sin\theta)}{5}$$

$$\sin\theta = \frac{3\sqrt{2}}{11}$$



$$v_1 = \sqrt{\frac{6gR(\sqrt{2}-2\sin\theta)}{5}}\cos\theta,\ v_2 = \sqrt{\frac{6gR(\sqrt{2}-2\sin\theta)}{5}}\sin\theta$$

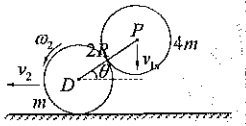
P碰到水平面时速度：

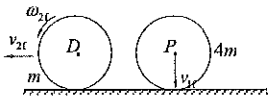


$$v_{1s} \equiv v_1\big|_{\sin\theta=\frac{3\sqrt{2}}{11}} = \frac{1}{11}\sqrt{\frac{618\sqrt{2}gR}{11}}$$

$$v_{1t} = \sqrt{v_{1s}^2 + 4gR\sin\theta}$$

$$= \frac{3}{11}\sqrt{\frac{230\sqrt{2}gR}{11}} \approx 1.48\sqrt{gR}$$





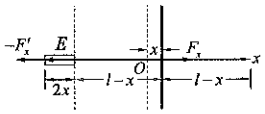
$$v_1 = \sqrt{\frac{6gR(\sqrt{2}-2\sin\theta)}{5}}\cos\theta$$

脱离位置：$\sin\theta = 3\sqrt{2}/11$

题四.

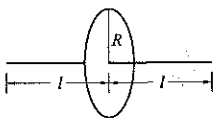
(1) $$E = \frac{1}{4\pi\varepsilon_0}\frac{Q(l-x)}{[(l-x)^2+R^2]^{3/2}}$$

$$\approx \frac{1}{4\pi\varepsilon_0}\frac{Ql}{(l^2+R^2)^{3/2}}$$



$$F_x = -F_x' = 2\lambda x\cdot E = \frac{qQ}{4\pi\varepsilon_0(l^2+R^2)^{3/2}}x \quad (q=\lambda\cdot 2l) \qquad F_x = ?$$

$$\equiv -k_1 x\left[k_1 = -\frac{qQ}{4\pi\varepsilon_0(l^2+R^2)^{3/2}}\right]$$



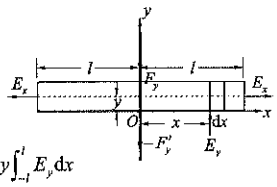
$qQ>0$: $k_1<0$, 不稳定平衡；

$qQ<0$: $k_1>0$, 稳定平衡。

$$T_1 = 2\pi\sqrt{\frac{m}{k_1}} = 4\pi\sqrt{\frac{\pi\varepsilon_0 m(l^2+R^2)^{3/2}}{|qQ|}}$$

(2)

$$\oint_S \vec{E}\cdot d\vec{S} = \int_{S_{侧}}\vec{E}\cdot d\vec{S} + \int_{S_{底}}\vec{E}\cdot d\vec{S}$$

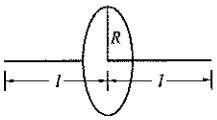


$$\int_{S_{底}}\vec{E}\cdot d\vec{S} \approx 2\pi y^2 E_x$$

$$\int_{S_{侧}}\vec{E}\cdot d\vec{S} = -\int_{-l}^{l}E_y\cdot 2\pi y dx = -2\pi y\int_{-l}^{l}E_y dx$$

$$\oint_S \vec{E}\cdot d\vec{S} = 2\pi y^2 E_x - 2\pi y\int_{-l}^{l}E_y dx$$

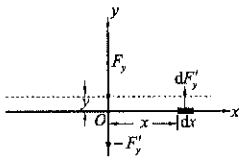
$$\oint_S \vec{E}\cdot d\vec{S} = 0 \Rightarrow$$



$$\int_{-l}^{l}E_y dx = E_x y = \frac{1}{4\pi\varepsilon_0}\frac{Ql}{(l^2+R^2)^{3/2}}y$$

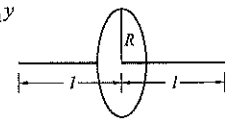
$$dF_y' = \lambda dx\cdot E_y = \lambda E_y dx$$

$$F_y' = \lambda\int_{-l}^{l}E_y dx = \frac{1}{4\pi\varepsilon_0}\frac{Q\lambda l}{(l^2+R^2)^{3/2}}y$$



$$= \frac{qQ}{8\pi\varepsilon_0(l^2+R^2)^{3/2}}y \quad (q=\lambda\cdot 2l)$$

$$F_y = -F_y' = -\frac{qQ}{8\pi\varepsilon_0(l^2+R^2)^{3/2}}y = -k_2 y$$



$qQ>0$: $k_2>0$, 稳定平衡；

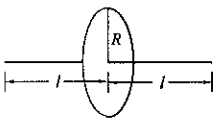
$qQ<0$: $k_2<0$, 不稳定平衡。

$$T_2 = 2\pi\sqrt{\frac{m}{k_2}} = 4\pi\sqrt{\frac{2\pi\varepsilon_0 m(l^2+R^2)^{3/2}}{qQ}}$$

$$\int_{-l}^{l}E_y dx = \frac{1}{4\pi\varepsilon_0}\frac{Ql}{(l^2+R^2)^{3/2}}y$$

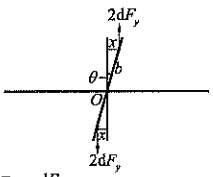
(3)

$$dF_r = E dq \approx \frac{\lambda}{2\pi\varepsilon_0 R}\frac{Q}{2\pi}d\varphi$$



$$dF_y = dF_r\cos\varphi \approx \frac{\lambda Q}{4\pi^2\varepsilon_0 R}\cos\varphi d\varphi$$

$$dM = -2\times 2dF_y\cdot x = -4dF_y\cdot b\theta$$

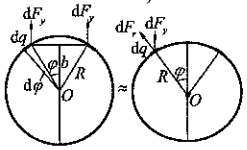


$$= -4dF_y R\cos\varphi\,\theta \approx -\frac{\lambda Q\theta}{\pi^2\varepsilon_0}\cos^2\varphi d\varphi$$

$$M = -\frac{\lambda Q\theta}{\pi^2\varepsilon_0}\int_{-\pi/2}^{\pi/2}\cos^2\varphi d\varphi = -\frac{\lambda Q}{2\pi\varepsilon_0}\theta$$

$$= -\frac{qQ}{4\pi\varepsilon_0 l}\theta \quad (q=\lambda\cdot 2l)$$

$$\equiv -k_3\theta \quad (k_3 = \frac{qQ}{4\pi\varepsilon_0 l})$$

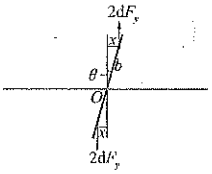


$qQ>0$: $k_3>0$, 稳定平衡;

$qQ<0$: $k_3<0$, 不稳定平衡。

$$J=\frac{1}{2}mR^2$$

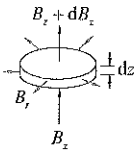
$$T=2\pi\sqrt{\frac{J}{k_3}}=2\pi R\sqrt{\frac{2\pi\varepsilon_0 ml}{qQ}}$$

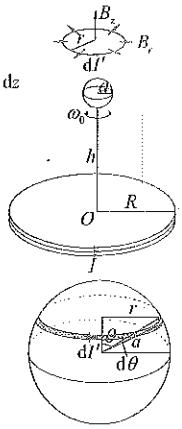


$$M=-k_3\theta\ (k_3=\frac{qQ}{4\pi\varepsilon_0 l})$$

题五.

(1) $B_z(z)=\dfrac{\mu_0}{2}\dfrac{R^2NI}{(z^2+R^2)^{3/2}}$





$$\oint_S \vec{B}\cdot d\vec{S}=0\Rightarrow$$

$$(B_z+dB_z)\cdot\pi r^2-B_z\cdot\pi r^2+B_r\cdot 2\pi r dz=0$$

$$B_r=-\frac{r}{2}\frac{dB_z}{dz}\bigg|_{z=h}=\frac{3\mu_0}{4}\frac{NIR^2hr}{(h^2+R^2)^{5/2}}$$

$$dF_z=B_r dl'\cdot 2\pi r$$

$$=\frac{3\pi\mu_0}{2}\frac{NIR^2hr^2}{(h^2+R^2)^{5/2}}dl'$$

$r=a\sin\theta,\ \sigma=\dfrac{Q}{4\pi a^2}$

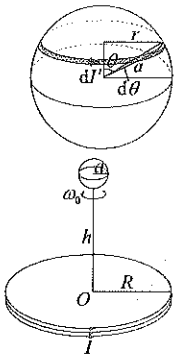
$dq=\sigma\cdot 2\pi r\cdot a d\theta=\dfrac{1}{2}Q\sin\theta d\theta,\ T=\dfrac{2\pi}{\omega_0}$

$$dI'=\frac{dq}{T}=\frac{Q\omega_0}{4\pi}\sin\theta d\theta$$

$$dF_z=\frac{3\mu_0}{8}\frac{Q\omega_0NIR^2ha^2}{(h^2+R^2)^{5/2}}\sin^3\theta d\theta$$

$$F_z=\frac{3\mu_0}{8}\frac{Q\omega_0NIR^2ha^2}{(h^2+R^2)^{5/2}}\int_0^\pi\sin^3\theta d\theta$$

$$=\frac{\mu_0}{2}\frac{Q\omega_0NIR^2ha^2}{(h^2+R^2)^{5/2}}$$



$$dF_z=\frac{3\pi\mu_0}{2}\frac{NIR^2hr^2}{(h^2+R^2)^{5/2}}dl'$$

另解：旋转的带电球壳可视为一个磁偶极子：$\mu=?$

$r=a\sin\theta,\ dI'=\dfrac{Q\omega_0}{4\pi}\sin\theta d\theta$

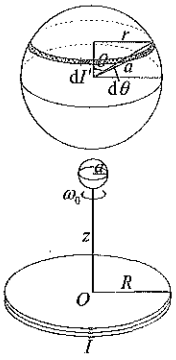
$$d\mu=dI'\cdot\pi r^2=\frac{Q\omega_0a^2}{4}\sin^3\theta d\theta$$

$$\mu=\frac{Q\omega_0a^2}{4}\int_0^\pi\sin^3\theta d\theta=\frac{Q\omega_0a^2}{3}$$

$$\vec{\mu}=-\mu\vec{e}_z$$

$$W_m=-\vec{\mu}\cdot\vec{B}=\mu B_z=\frac{\mu_0}{6}\frac{Q\omega_0NIR^2a^2}{(z^2+R^2)^{3/2}}$$

$$F_z=-\frac{\partial W_m}{\partial z}\bigg|_{z=h}=\frac{\mu_0}{2}\frac{Q\omega_0NIR^2ha^2}{(h^2+R^2)^{5/2}}$$



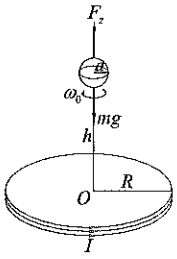
$$dI'=\frac{Q\omega_0}{4\pi}\sin\theta d\theta,\ B_z(z)=\frac{\mu_0}{2}\frac{R^2NI}{(z^2+R^2)^{3/2}}$$

$F_z=mg\Rightarrow$

$$\frac{\mu_0}{2}\frac{Q\omega_0NIR^2ha^2}{(h^2+R^2)^{5/2}}=mg$$

$$\omega_0=\frac{2mg(h^2+R^2)^{5/2}}{\mu_0QNIR^2ha^2}$$

方向：顺时针（从上方看）



$$F_z=\frac{\mu_0}{2}\frac{Q\omega_0NIR^2ha^2}{(h^2+R^2)^{5/2}}$$

(2)

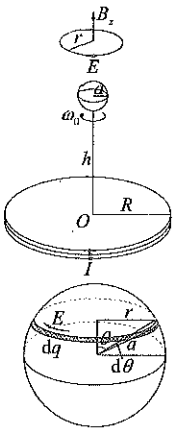
$$\oint_L\vec{E}\cdot d\vec{l}=E\cdot 2\pi r,\quad \Phi=-B_z\cdot\pi r^2$$

$$\oint_L\vec{E}\cdot d\vec{l}=-\frac{d\Phi}{dt}\Rightarrow E=\frac{r}{2}\frac{dB_z}{dt}$$

$$d\tau=r\cdot dqE=\frac{1}{4}Q\frac{dB_z}{dt}r^2\sin\theta d\theta$$

$$=\frac{1}{4}Qa^2\frac{dB_z}{dt}\sin^3\theta d\theta\quad(r=a\sin\theta)$$

$$=\frac{\mu_0}{8}\frac{QNR^2a^2}{(h^2+R^2)^{3/2}}\frac{dI}{dt}\sin^3\theta d\theta$$



$$B_z(h)=\frac{\mu_0}{2}\frac{R^2NI}{(h^2+R^2)^{3/2}},\ dq=\frac{1}{2}Q\sin\theta d\theta$$

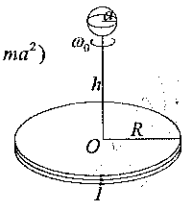
$$\tau=\frac{\mu_0}{8}\frac{QNR^2a^2}{(h^2+R^2)^{3/2}}\frac{\mathrm{d}I}{\mathrm{d}t}\int_0^{\pi}\sin^3\theta\mathrm{d}\theta$$

$$=\frac{\mu_0}{6}\frac{QNR^2a^2}{(h^2+R^2)^{3/2}}\frac{\mathrm{d}I}{\mathrm{d}t}=\frac{2}{3}ma^2\frac{\mathrm{d}\omega}{\mathrm{d}t}\quad(J=\frac{2}{3}ma^2)$$

$$\mathrm{d}\omega=\frac{\mu_0}{4}\frac{QNR^2}{m(h^2+R^2)^{3/2}}\mathrm{d}I$$

$$\int_{\omega_0}^{\omega}\mathrm{d}\omega=\frac{\mu_0}{4}\frac{QNR^2}{m(h^2+R^2)^{3/2}}\int_I^0\mathrm{d}I$$

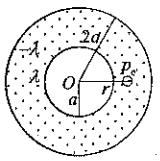
$$\omega=\omega_0-\frac{\mu_0}{4}\frac{QNIR^2}{m(h^2+R^2)^{3/2}}$$



$$B_z(h)=\frac{\mu_0}{2}\frac{R^2NI}{(h^2+R^2)^{3/2}},\mathrm{d}q=\frac{1}{2}Q\sin\theta\mathrm{d}\theta$$

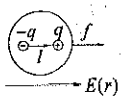
题六.

(1) $\begin{cases}E(r)=\dfrac{\lambda}{2\pi\varepsilon_0 r}\\ V=\displaystyle\int_a^{2a}E\mathrm{d}r=\int_a^{2a}\frac{\lambda}{2\pi\varepsilon_0 r}\mathrm{d}r=\frac{\lambda}{2\pi\varepsilon_0}\ln 2\end{cases}$



$$E(r)=\frac{V}{r\ln 2}$$

$$p_e=\gamma E=\frac{\gamma V}{r\ln 2}$$



$$f=qE(r+l)-qE(r)\approx q\frac{\mathrm{d}E}{\mathrm{d}r}l\approx p_e\frac{\mathrm{d}E}{\mathrm{d}r}=-\frac{\gamma V^2}{r^3(\ln 2)^2}$$

(2) 延拓：$a<r<2a\Rightarrow a<r<\infty$

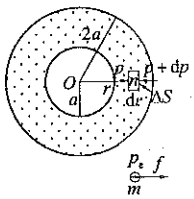
$$f=-\frac{\gamma U^2}{r^3(\ln 2)^2}$$

$w(\infty)=0$

$$w(r)=\int_r^{\infty}f\mathrm{d}r=-\frac{\gamma V^2}{(\ln 2)^2}\int_r^{\infty}\frac{1}{r^3}\mathrm{d}r=w_a\frac{a^2}{r^2}\quad\left[w_a=-\frac{\gamma V^2}{2a^2(\ln 2)^2}\right]$$

$$p\Delta S+f\cdot n\cdot\Delta S\mathrm{d}r=(p+\mathrm{d}p)\Delta S$$

$$\left.\begin{aligned}\mathrm{d}p&=nf\mathrm{d}r\\ p&=nkT\end{aligned}\right\}\Rightarrow kT\mathrm{d}n=nf\mathrm{d}r$$



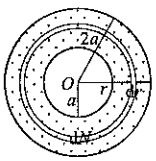
$$\frac{\mathrm{d}n}{n}\approx\frac{1}{kT}f\mathrm{d}r$$

$$\int_n^{n_\infty}\frac{\mathrm{d}n}{n}=\frac{1}{kT}\int_r^{\infty}f\mathrm{d}r=\frac{w(r)}{kT}$$

$$n(r)=n_\infty\exp[-\frac{w(r)}{kT}]=n_\infty\exp(-\frac{w_a}{kT}\frac{a^2}{r^2})\quad[\exp(x)=\mathrm{e}^x]$$

$$\mathrm{d}N=n\cdot 2\pi lr\mathrm{d}r=2\pi ln_\infty\exp(-\frac{w_a}{kT}\frac{a^2}{r^2})r\mathrm{d}r$$

$$N=2\pi ln_\infty\int_a^{2a}\exp(-\frac{w_a}{kT}\frac{a^2}{r^2})r\mathrm{d}r$$



$$t=\frac{a^2}{r^2}\Rightarrow r=\frac{a}{\sqrt{t}}$$

$$N=-\pi a^2ln_\infty\int_1^{1/4}\exp(-\frac{w_a}{kT}t)t^{-2}\mathrm{d}t=\pi a^2ln_\infty I_2(\frac{w_a}{kT})$$

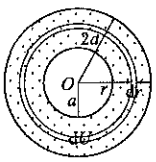
$$n_\infty=\frac{N}{\pi a^2lI_2(w_a/kT)}$$

$$n(r)=\frac{N}{\pi a^2lI_2(w_a/kT)}\exp(-\frac{w_a}{kT}\frac{a^2}{r^2})\quad\left[w_a=-\frac{\gamma V^2}{2a^2(\ln 2)^2}\right]$$

$$n(r)=n_\infty\exp(\frac{w_a}{kT}\frac{a^2}{r^2}),I_n(x)=\int_{1/4}^1 e^{-xt}t^{-n}\mathrm{d}t$$

(3)

$$u=\frac{5}{2}kT+w(r)=\frac{5}{2}kT+w_a\frac{a^2}{r^2}$$



$$\mathrm{d}U=u\mathrm{d}N=\frac{5}{2}kT\mathrm{d}N+w_a\frac{a^2}{r^2}\mathrm{d}N$$

$$U=\frac{5}{2}NkT+w_aa^2\int\frac{1}{r^2}\mathrm{d}N$$

$$=\frac{5}{2}NkT+2\pi a^2lw_an_\infty\int_a^{2a}\exp(-\frac{w_a}{kT}\frac{a^2}{r^2})\frac{1}{r}\mathrm{d}r$$

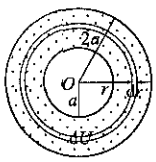
$$t=a^2/r^2\Rightarrow r=a/\sqrt{t}$$

$$U=\frac{5}{2}NkT+\pi a^2lw_an_\infty\int_{1/4}^1\exp(-\frac{w_a}{kT}t)t^{-1}\mathrm{d}t=\frac{5}{2}NkT+Nw_a\frac{I_1(w_a/kT)}{I_2(w_a/kT)}$$

$$w(r)=w_a\frac{a^2}{r^2},\mathrm{d}N=2\pi ln_\infty\exp(-\frac{w_a}{kT}\frac{a^2}{r^2})r\mathrm{d}r,n_\infty=\frac{N}{\pi a^2lI_2(w_a/kT)}$$

$$C_V=\frac{\mathrm{d}U}{\mathrm{d}T}=\frac{5}{2}Nk+Nw_a\frac{\mathrm{d}}{\mathrm{d}T}\frac{I_1(w_a/kT)}{I_2(w_a/kT)}$$

$$x=w_a/kT\Rightarrow$$



$$C_V=\frac{5}{2}Nk+Nw_a\frac{\mathrm{d}}{\mathrm{d}x}(\frac{I_1(x)}{I_2(x)})\cdot\frac{\mathrm{d}x}{\mathrm{d}t}$$

$$=\frac{5}{2}Nk-Nk(\frac{w_a}{kT})^2\frac{I_1^2(x)-I_0(x)I_2(x)}{I_2^2(x)}$$

$$=Nk\left\{\frac{5}{2}-(\frac{w_a}{kT})^2\left[\frac{I_1^2(w_a/kT)}{I_2^2(w_a/kT)}-\frac{I_0(w_a/kT)}{I_2(w_a/kT)}\right]\right\}$$

$$\left[I_n(x)=\int_{1/4}^1\mathrm{e}^{-xt}t^{-n}\mathrm{d}t,w_a=-\frac{\gamma V^2}{2a^2(\ln 2)^2}\right]$$

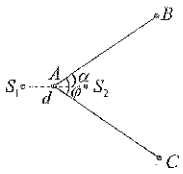
$$U=\frac{5}{2}NkT+Nw_a\frac{I_1(w_a/kT)}{I_2(w_a/kT)},\frac{\mathrm{d}I_n(x)}{\mathrm{d}x}=-I_{n-1}(x)$$

题七.

(1) $\varphi_{S_1}-\varphi_{S_2}=\Delta\varphi_0$, $\overline{S_1B}-\overline{S_2B}\approx d\cos\alpha$

$$\Delta\varphi_B=\varphi_{S_1}-\varphi_{S_2}=\frac{2\pi(\overline{S_1B}-\overline{S_2B})}{\lambda}$$

$$=\Delta\varphi_0-\frac{2\pi d\cos\alpha}{\lambda}$$

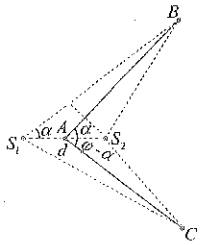


同理:

$$\Delta\varphi_C=\Delta\varphi_0-\frac{2\pi d\cos(\varphi-\alpha)}{\lambda}$$

$$\Delta\varphi_B-\Delta\varphi_C=\frac{2\pi d}{\lambda}[\cos(\varphi-\alpha)-\cos\alpha]$$

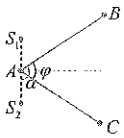
$$=\frac{4\pi d}{\lambda}\sin\frac{\varphi}{2}\sin(\alpha-\frac{\varphi}{2})$$



$$\Delta\varphi_B-\Delta\varphi_C=\frac{4\pi d}{\lambda}\sin\frac{\varphi}{2}\sin(\alpha-\frac{\varphi}{2})=(2k+1)\pi$$

$$d=\frac{2k+1}{4}\frac{\lambda}{\sin(\varphi/2)\sin(\alpha-\varphi/2)}$$

$$d=\left|\frac{2k+1}{4}\right|\frac{\lambda}{|\sin(\varphi/2)||\sin(\alpha-\varphi/2)|}$$



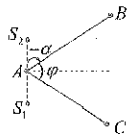
$$\alpha=\frac{\varphi+\pi}{2}$$

$k=0,\sin(\alpha-\frac{\varphi}{2})=1,\ \alpha-\frac{\varphi}{2}=\frac{\pi}{2}\Rightarrow\alpha=\frac{\varphi+\pi}{2}$

$(-3\pi/2\le\alpha-\varphi/2\le\pi)$

或

$k=-1,\sin(\alpha-\frac{\varphi}{2})=-1,\ \alpha-\frac{\varphi}{2}=-\frac{\pi}{2}\Rightarrow\alpha=\frac{\varphi-\pi}{2}$



$$\alpha=\frac{\varphi-\pi}{2}$$

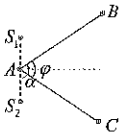
$$d_{min}=\frac{\lambda}{4\sin(\varphi/2)}$$

$0\le\varphi\le\pi,-\pi\le\alpha\le\pi$

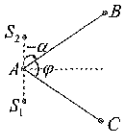
(2)

$$d=d_{min}=\frac{\lambda}{4\sin(\varphi/2)},\ \alpha=\frac{\varphi\pm\pi}{2}$$

$$\Delta\varphi_B=\begin{cases}\Delta\varphi_0+\frac{\pi}{2}, & \alpha=\frac{\varphi+\pi}{2}\\ \Delta\varphi_0-\frac{\pi}{2}, & \alpha=\frac{\varphi-\pi}{2}\end{cases}$$



$$\alpha=\frac{\varphi+\pi}{2}$$



$$\alpha=\frac{\varphi-\pi}{2}$$

$\Delta\varphi_B=2k\pi\Rightarrow$

$$\Delta\varphi_0=\begin{cases}2k\pi-\frac{\pi}{2}, & \alpha=\frac{\varphi+\pi}{2}\\ 2k\pi+\frac{\pi}{2}, & \alpha=\frac{\varphi-\pi}{2}\end{cases}$$

$$\Delta\varphi_B=\Delta\varphi_0-\frac{2\pi d\cos\alpha}{\lambda}$$

$$d=d_{min}=\frac{\lambda}{4\sin(\varphi/2)}$$

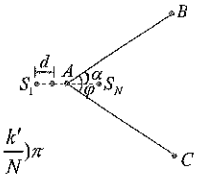
(3)

$$\Delta\varphi_B=\Delta\varphi_0-\frac{2\pi d\cos\alpha}{\lambda}$$

$$\Delta\varphi_C=\Delta\varphi_0-\frac{2\pi d\cos(\varphi-\alpha)}{\lambda}$$

$$\Delta\varphi_B-\Delta\varphi_C=\frac{4\pi d}{\lambda}\sin\frac{\varphi}{2}\sin(\alpha-\frac{\varphi}{2})=2(k+\frac{k'}{N})\pi$$

$[k'=\pm1,\pm2,\cdots,\pm(N-1)]$

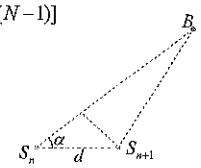


$$d=\frac{k+k'/N}{2}\frac{\lambda}{\sin(\varphi/2)\sin(\alpha-\varphi/2)}$$

$k=0,k'=1,\alpha-\frac{\varphi}{2}=\frac{\pi}{2}\Rightarrow\alpha=\frac{\varphi+\pi}{2}$

或

$k=0,k'=-1,\alpha-\frac{\varphi}{2}=-\frac{\pi}{2}\Rightarrow\alpha=\frac{\varphi-\pi}{2}$



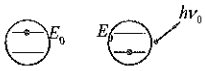
$$d_{min}=\frac{\lambda}{2N\sin(\varphi/2)}$$

多光束干涉暗纹中心:

$\delta=\pm2(k+\frac{k'}{N})\pi\ (k'=1,\cdots N-1)$

题八.

(1) $h\nu_0=E_0$

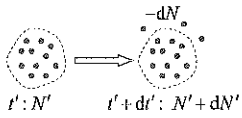


以v运动的参照系S'

$$-\mathrm{d}N'=\lambda N'\mathrm{d}t'$$

$$-\frac{\mathrm{d}N'}{N'}=\lambda\mathrm{d}t'$$

$$-\int_{N_0}^{N'}\frac{\mathrm{d}N'}{N'}=\lambda\int_0^{t'}\mathrm{d}t'$$

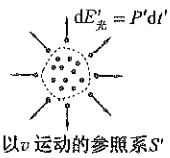


$$N'=N_0'\mathrm{e}^{-\lambda t'},\ \dot{N}'=\frac{\mathrm{d}N'}{\mathrm{d}t'}=-\lambda N_0\mathrm{e}^{-\lambda t'}$$

$$P'=-\dot{N}h\nu_0=\lambda N_0E_0\mathrm{e}^{-\lambda t'}$$

(2)

$$\begin{cases}\mathrm{d}E'_{光}=P'\mathrm{d}t'\\ \mathrm{d}E_{光}=P\mathrm{d}t\\ \mathrm{d}t=\gamma\mathrm{d}t'\ \ (\gamma=1/\sqrt{1-\beta^2},\beta=v/c)\\ \mathrm{d}E_{光}=\gamma(\mathrm{d}E'_{光}+v\mathrm{d}p'_{光x})=\gamma\mathrm{d}E'_{光}\end{cases}$$



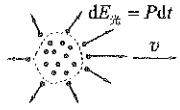
以v运动的参照系S'

$$P=P'=\lambda N_0E_0\mathrm{e}^{-\lambda t'}$$

$$t=\gamma t'$$

$$P=\lambda N_0E_0\mathrm{e}^{-\lambda t/\gamma}$$

$$(\gamma=1/\sqrt{1-\beta^2},\beta=v/c)$$



实验室参照系S

$$P'=\lambda N_0E_0\mathrm{e}^{-\lambda t'}$$

(3)

$$\mathrm{d}\dot{N}'\mathrm{d}t' = \dot{n}'\mathrm{d}t'\cdot 2\pi\sin\theta'\mathrm{d}\theta' = \frac{\dot{N}'}{4\pi}\mathrm{d}t'\cdot 2\pi\sin\theta'\mathrm{d}\theta'$$

($\dot{n}'$: S'系单位时间从单位立体角出射的光子数)

$$\mathrm{d}\dot{N}\mathrm{d}t = \dot{n}\mathrm{d}t\cdot 2\pi\sin\theta\mathrm{d}\theta$$

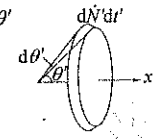
$$\mathrm{d}\dot{N}\mathrm{d}t = \mathrm{d}\dot{N}'\mathrm{d}t' \Rightarrow \dot{n} = \frac{\dot{N}'}{4\pi}\frac{\sin\theta'\mathrm{d}\theta'}{\sin\theta\mathrm{d}\theta}\frac{\mathrm{d}t'}{\mathrm{d}t} \quad (1)$$

$$\cos\theta' = \frac{\cos\theta-\beta}{1-\beta\cos\theta}$$

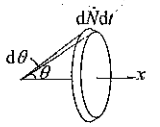
$$-\sin\theta'\mathrm{d}\theta' = -\frac{(1-\beta^2)\sin\theta\mathrm{d}\theta}{(1-\beta\cos\theta)^2} \quad (2)$$

$$\mathrm{d}t' = \mathrm{d}t\sqrt{1-\beta^2} \quad (3)$$

$$(1)(2) \Rightarrow \dot{n} = \frac{\dot{N}'}{4\pi}\frac{(1-\beta^2)^{3/2}}{(1-\beta\cos\theta)^2}$$



以v运动的参照系S'



实验室参照系S

$$\dot{n} = \frac{\dot{N}'}{4\pi}\frac{(1-\beta^2)^{3/2}}{(1-\beta\cos\theta)^2},\ \nu = \frac{\nu_0\sqrt{1-\beta^2}}{1-\beta\cos\theta}$$

$$\dot{w} = \frac{\mathrm{d}P}{\mathrm{d}\Omega} = \dot{n}h\nu = \frac{\dot{N}'h\nu_0}{4\pi}\frac{(1-\beta^2)^2}{(1-\beta\cos\theta)^3}$$

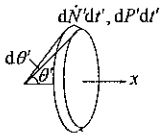
$$= \frac{P'}{4\pi}\frac{(1-\beta^2)^2}{(1-\beta\cos\theta)^3}$$

$$= \frac{P}{4\pi}\frac{(1-\beta^2)^2}{(1-\beta\cos\theta)^3} \quad (P = \lambda N_0 E_0 \mathrm{e}^{-\lambda t/\gamma})$$

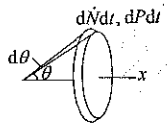
验证：

$$\int_0^\pi \dot{w}\cdot 2\pi\sin\theta\mathrm{d}\theta$$

$$= \frac{1}{2}P(1-\beta^2)^2\int_0^\pi\frac{\sin\theta\mathrm{d}\theta}{(1-\beta\cos\theta)^3} = P$$



以v运动的参照系S'



实验室参照系S

$$\dot{n} = \frac{\dot{N}'}{4\pi}\frac{(1-\beta^2)^{3/2}}{(1-\beta\cos\theta)^2}$$

$$P' = P,\ P = \lambda N_0 E_0 \mathrm{e}^{-\lambda t/\gamma}$$

(3) 参照系S'：

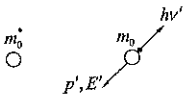
$$\begin{cases} E' + h\nu' = m_0^* c^2 \\ p' = h\nu'/c \\ E'^2 = (m_0c^2)^2 + p'^2c^2 \\ m_0^*c^2 = m_0c^2 + E_0 \end{cases}$$

$$h\nu' = E_0\frac{E_0+2m_0c^2}{2E_0+2m_0c^2}$$

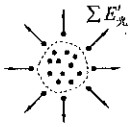
N_0个原子H发出的光子的总能量：

$$\sum E'_{光} = N_0h\nu' = N_0E_0\frac{E_0+2m_0c^2}{2E_0+2m_0c^2}$$

总动量：$\sum \vec{p}'_{光} = 0$



以v运动的参照系S'



以v运动的参照系S'

实验室参照系S：

N_0个原子H发出的光子的总能量：

$$\sum E_{光} = \gamma(\sum E'_{光} + v\sum p'_{yx}) = \gamma\sum E'_{光}$$

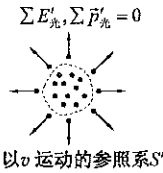
$$= \gamma N_0E_0\frac{E_0+2m_0c^2}{2E_0+2m_0c^2}$$

总动量：

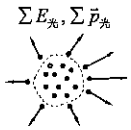
$$\sum p_{光} = \gamma(\sum p'_{光x} + \frac{v}{c^2}\sum E'_{光x}) = \frac{\gamma v}{c^2}\sum E'_{光x}$$

$$= \frac{\gamma\beta N_0E_0}{c}\frac{E_0+2m_0c^2}{2E_0+2m_0c^2}$$

$$\sum E'_{光} = N_0E_0\frac{E_0+2m_0c^2}{2E_0+2m_0c^2},\ \sum \vec{p}'_{光} = 0$$



以v运动的参照系S'



实验室参照系S